

USED CAR MARKET ANALYTICS

CARVISTA

A Multi-Page Power BI Dashboard for the Used Car Market

Project Documentation

Prepared by Pathan Tarannum Bilal

Power BI | DAX | Power Query | Data Science Learning Journey

Table of Contents

01 Project Overview.....03

02 Complete Dashboard Overview.....04

03 Dashboard Page 1 — Overview.....05

04 Dashboard Page 2 — Pricing Analysis.....06

05 Dashboard Page 3 — Specifications.....07

06 Dashboard Page 4 — Market Insights.....08

07 Key Findings.....09

08 Skills Demonstrated.....10

09 Conclusion.....10

01 Project Overview

CARVISTA is a multi-page Power BI dashboard built to analyze the Indian used-car market. The project goes beyond simply charting numbers — it is designed around the question every good BI report should answer: what business decision does this visual support? The dashboard covers pricing behaviour, technical specifications, and broader market trends across brands, fuel types, and vehicle age, all backed by a fully interactive filtering system.

Objective

- Turn a raw used-car dataset into a decision-ready analytics tool for buyers, sellers, and dealers.
- Practice the full BI workflow: data cleaning, modelling, DAX measures, and dashboard UX design.
- Build a hands-on project as part of an ongoing data science learning journey.

Tools & Technologies

Power BI Dashboard & Visuals	Power Query Data Cleaning (ETL)	DAX Measures & KPIs	Excel Supporting Analysis
--	---	-------------------------------	-------------------------------------

Dataset Snapshot

- 15,244 used-car listings analyzed across 31 active brands and 119 models.
- Fields include brand, model, price, kilometers driven, fuel type, transmission, seller type, vehicle age, mileage, engine capacity, power, and seating capacity.
- Average selling price: ₹7.75 Lakh | Average vehicle age: 6.04 years | Average mileage: 19.7 km/l.

02 Complete Dashboard Overview

CARVISTA uses a dark navy-and-purple theme with pink accents to give the report a premium, automotive feel while keeping every visual easy to read. The layout stays consistent across all four pages: a left-hand filter panel (Brand, Model, Fuel Type, Transmission, Seller Type, Vehicle Age), a top navigation bar to switch pages, a car-image panel that updates dynamically with the selected brand/model, and a row of KPI cards for instant context.

Every page is built around a different business question — pricing behaviour, technical specifications, or market-level trends — but all four stay synchronized: changing a filter on one page updates every page's KPIs, charts, and insight cards together. This is what turns the report from a set of static charts into a single connected analytical tool.

Key Metrics: 15,244 cars analyzed · 7.75 Lakh avg selling price · 55,640 avg KM driven · 6.04 yrs avg vehicle age · 19.70 km/l ₹ avg mileage

How the Four Pages Connect

- Overview — market snapshot: totals, top brands, fuel/transmission split, and price-vs-age trend.
- Pricing Analysis — how price varies by brand, fuel type, transmission, and usage (KM driven).
- Specifications — the technical side: power, mileage, engine size, and seat configuration.
- Market Insights — brand share, seller-type behaviour, and mileage/power trends at a market level.

03 Dashboard Walkthrough

DASHBOARD PAGE 1

Overview



The Overview page gives a single-glance summary of the entire used-car market. KPI cards up top surface total listings, average selling price, distance driven, and vehicle age, while the charts below break down the top brands by volume and the fuel/transmission mix — everything needed to understand the dataset before drilling into specifics.

Key Metrics: 15,244 total cars · 774.70K avg price · 55,640 avg KM driven · 6.04 yrs avg age · 19.70 km/l avg mileage₹

Key Insights

- Maruti leads the market with 4,933 cars (32.4% share).
- Petrol dominates the fuel mix at 49.6% of listings.

-
- Selling price drops sharply as vehicle age increases — from ₹1.14M (0–2 yrs) to ₹0.33M (10+ yrs).
 - Manual transmission accounts for 79.3% of listings vs. 20.7% automatic.

Pricing Analysis



This page breaks down average selling price by brand, fuel type, and transmission, alongside a KM-driven vs. price scatter and a vehicle-age vs. price trend. It's built to answer the question a buyer or dealer actually cares about: what drives the price up or down?

Key Metrics: 15,244 cars · 774.70K avg price · 55,640 avg KM driven · 6.04 yrs avg age ₹

Key Insights

- Luxury brands (Ferrari, Rolls-Royce, Bentley) command the highest average prices, up to ₹39.5M.
- Electric vehicles have the highest average price by fuel type at ₹1.85M.
- Automatic transmission cars make up only 26.4% of listings but carry a clear price premium over manual.
- Price is highest for cars aged 0–2 years at ₹11.38 Lakh, then declines steadily with age.

Specifications



The Specifications page moves from price to performance — power distribution, mileage by fuel type, seat configuration, and engine category. It's aimed at buyers comparing cars on efficiency and capability rather than price alone.

Key Metrics: 19.70 avg mileage · 1.49K avg engine (cc) · 100.61 avg power (bhp) · 5-Seater top configuration

Key Insights

- Moderate-power cars dominate with 8,800 listings; performance-tier cars are rare at just 0.2K.
- CNG vehicles deliver the highest average mileage at 25.82 km/l.
- 5-seater cars are the most common configuration, at 83.8% share.
- Cars with medium-to-very-large engine categories make up over 70% of the market.

Market Insights



The final page zooms out to market-level dynamics — mileage distribution, engine category share, mileage by fuel type, and a power-vs-price scatter — rounding out the story with seller behaviour and broader trends across the market.

Key Metrics: 15K total cars · 119 active models · 31 active brands · ₹774.70K avg price · 19.70 avg mileage

Key Insights

- Dealer-sellers dominate the selected market with 62.1% share.
- Electric cars have the highest average selling price at ₹18.54 Lakh.
- CNG vehicles are, on average, the newest on the road at 5.5 years old.
- Maruti leads the market with 4,933 cars (32.4% share), consistent with the Overview page.

04 Key Findings

Pulling insights together across all four pages, a few patterns stand out consistently in the data:

- Brand concentration: Maruti and Hyundai together account for a large share of the market (32.4% and 19.3% respectively), so budget-segment demand drives most listings.
- Depreciation is steep and steady: average price nearly quadruples between 10+-year-old cars (₹0.33M) and 0–2-year-old cars (₹1.14M).
- Fuel type shapes price more than brand alone: Electric vehicles top the price charts by fuel type, even though they're a small share of total listings.
- Efficiency vs. price trade-off: CNG cars are both the most fuel-efficient (25.82 km/l) and among the more affordable options — a clear value segment.
- Dealers, not individuals, dominate supply: 62.1% of listings come through dealer sellers, which affects pricing consistency and negotiation room.

05 Skills Demonstrated

- **Power Query:** Data Cleaning & Transformation
- Calculated columns and dynamic DAX measures for KPIs and insight cards.
- Cross-page synchronized slicers for a seamless, connected user experience.
- Dynamic imagery (car images updating with brand/model selection) via conditional logic.
- Dashboard UX/UI design — consistent theme, layout hierarchy, and readability across four pages.
- Translating raw tabular data into business-relevant insight narratives, not just charts.

06 Conclusion

CARVISTA started as a way to practice Power BI end-to-end — from cleaning raw data to designing a dashboard someone could actually use to make a decision. The biggest takeaway wasn't a DAX formula; it was a mindset shift: a dashboard isn't judged by how many visuals it has, but by how clearly each one answers a real question. Connecting slicers, measures, and dynamic insight cards into one coherent story across four pages is what turned this into a genuine analytical tool rather than a slide of charts — and it's a milestone in an ongoing data science learning journey.

CARVISTA — Used Car Market Analytics

Built by Pathan Tarannum | Power BI Project — Data Science Learning Journey